

PORTFOLIO OF EVIDENCE REPORT

FORMATIVE ASSESSMENT 3 (FA3 POE): CAMPUS NETWORK INFRASTRUCTURE

Module Title: Networks Module

Code: NET511

Student Name and Surname: Tshebetso Cullil Mogashane

Student Number: 20261947

Institution: CTU Training Solutions

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Examiner: Linda Dube Internal

Moderator: Onias Sayi Internal

Moderator: Faith Muwishi and Palesa Nkosi

DECLARATION OF AUTHENTICITY AND RULES AND REGULATIONS - THEORY

I Tshebetso Cullil Mogashane

(FULL NAME)

hereby declare that the contents of this assessment

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is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.

Signature: _____

Date: 26/06/2026

Table of Contents

1. Introduction
2. Task 1 – Basic Network
3. Task 2 – IP Addressing and Subnetting
4. Task 3 – VLAN Configuration
5. Task 4 – Inter-VLAN Routing
6. Task 5 – Final Network Project
7. Conclusion
8. References
9. Declaration of Authenticity

Introduction

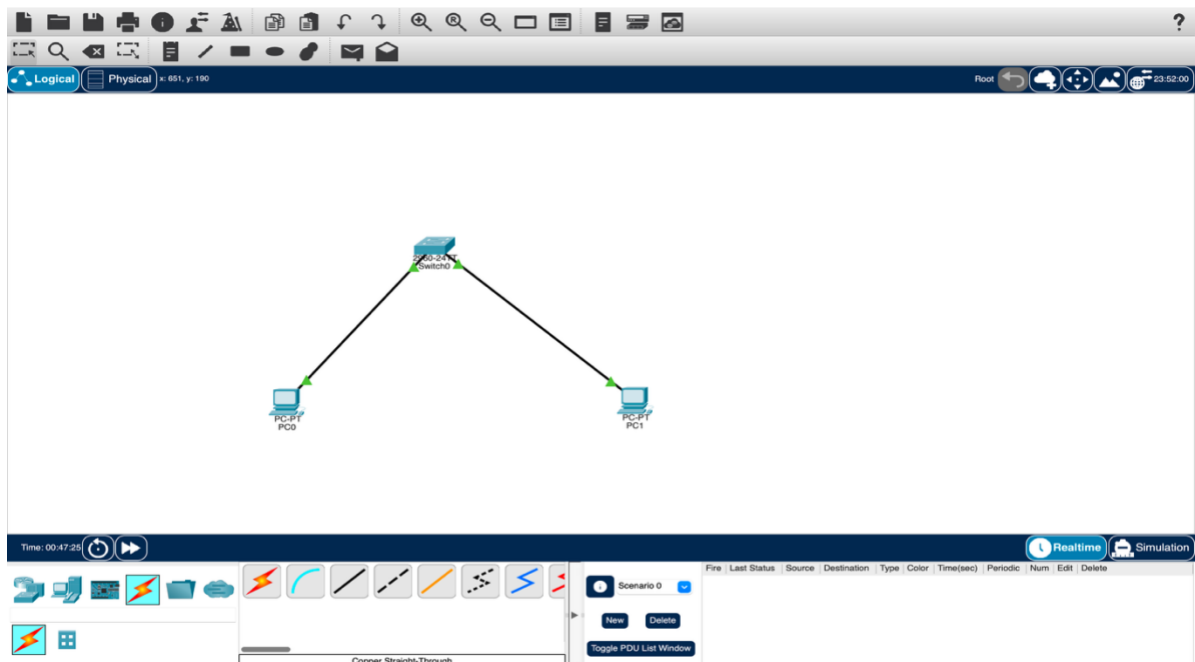
This practical assessment was completed using Cisco Packet Tracer to develop networking skills through the design and implementation of different network configurations. The project covered network fundamentals, IP addressing, subnetting, VLAN configuration, and inter-VLAN routing. Each task built upon the previous one, allowing the final project to demonstrate a fully functioning network with proper communication between multiple VLANs.

1. Task 1 – Basic Network

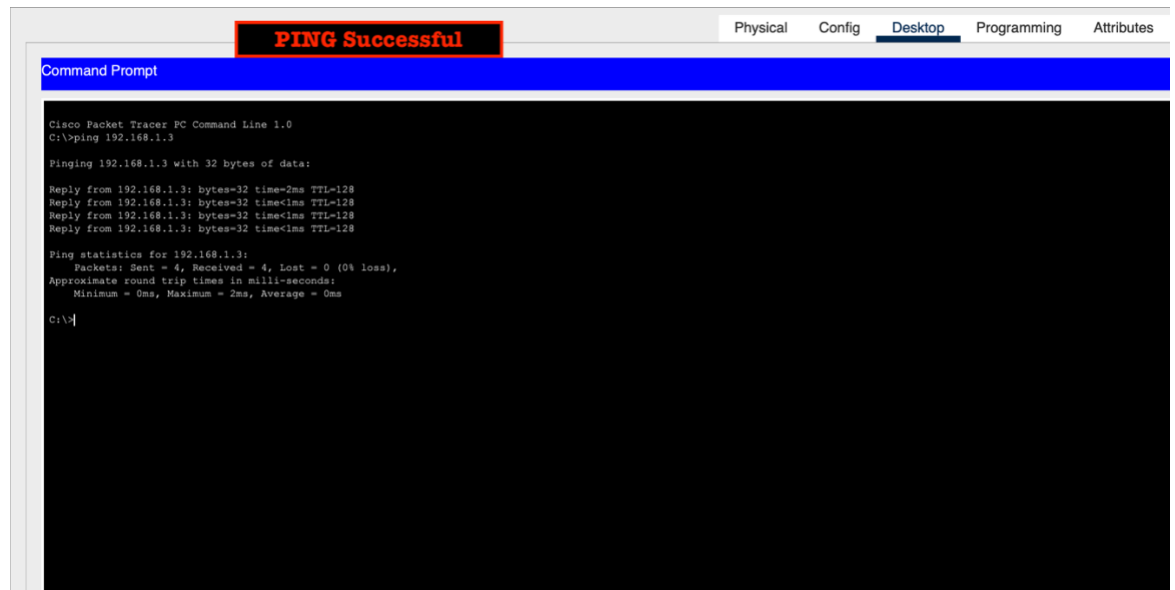
- A basic network created using a layer 2 switch and two devices with each device connected to the switch using a straight through cable and the static IP addresses assigned manually. The connectivity between the devices was verified using a successful ping test.

- **Evidence**

- **Figure 1: Network Topology**



- **Figure 2: Successful Ping Test**



2. Task 2 – IP Addressing and Subnetting

- In this task I simply expanded the network by adding two additional devices to the network and addressed them within the 192.168.10.0/24 network.
- Then I divided that network into two subnets using subnetting from /24 to /25 subnets which are subnet 1- 192.168.10.0/25 and subnet 2- 192.168.10.128/25 demonstrating how one network can be divided into two sub networks by using host bits.

- **Evidence**

- **Figure 3: IP Configuration Before subnetting**

PC 4

Physical	Config	Desktop	Programming	Attributes
----------	--------	----------------	-------------	------------

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.10.4

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

PC 3

Physical	Config	Desktop	Programming	Attributes
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IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.10.3

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

PC 2

Physical	Config	Desktop	Programming	Attributes
----------	--------	----------------	-------------	------------

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.10.2

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

PC 1

Physical	Config	Desktop	Programming	Attributes
IP Configuration				
Interface: FastEthernet0				
IP Configuration				
☐ DHCP ☒ Static				
IPv4 Address: 192.168.10.1				
Subnet Mask: 255.255.255.0				
Default Gateway: 0.0.0.0				
DNS Server: 0.0.0.0				

- **Figure 4: Ping Tests**

PC4 IP192.168.10.4(PING to PC1,PC2,PC3	Physical	Config	Desktop	Programming	Attributes
Command Prompt					
<pre>Cisco Packet Tracer PC Command Line 1.0 C:\>ping 192.168.10.1 Pinging 192.168.10.1 with 32 bytes of data: Reply from 192.168.10.1: bytes=32 time<1ms TTL=128 Reply from 192.168.10.1: bytes=32 time<1ms TTL=128 Reply from 192.168.10.1: bytes=32 time<1ms TTL=128 Reply from 192.168.10.1: bytes=32 time<1ms TTL=128 Ping statistics for 192.168.10.1: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 2ms, Average = 0ms C:\>ping 192.168.10.2 Pinging 192.168.10.2 with 32 bytes of data: Reply from 192.168.10.2: bytes=32 time<1ms TTL=128 Reply from 192.168.10.2: bytes=32 time<1ms TTL=128 Reply from 192.168.10.2: bytes=32 time<1ms TTL=128 Reply from 192.168.10.2: bytes=32 time<1ms TTL=128 Ping statistics for 192.168.10.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms C:\>ping 192.168.10.3 Pinging 192.168.10.3 with 32 bytes of data: Reply from 192.168.10.3: bytes=32 time<1ms TTL=128 Reply from 192.168.10.3: bytes=32 time<1ms TTL=128 Reply from 192.168.10.3: bytes=32 time<1ms TTL=128 Reply from 192.168.10.3: bytes=32 time<1ms TTL=128 Ping statistics for 192.168.10.3: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms</pre>					

PC3 IP192.168.10.3(PING to PC1,PC2,PC4

Physical

Config

Desktop

Programming

Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC2 IP192.168.10.2(PING to PC1,PC3,PC4

Physical

Config

Desktop

Programming

Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

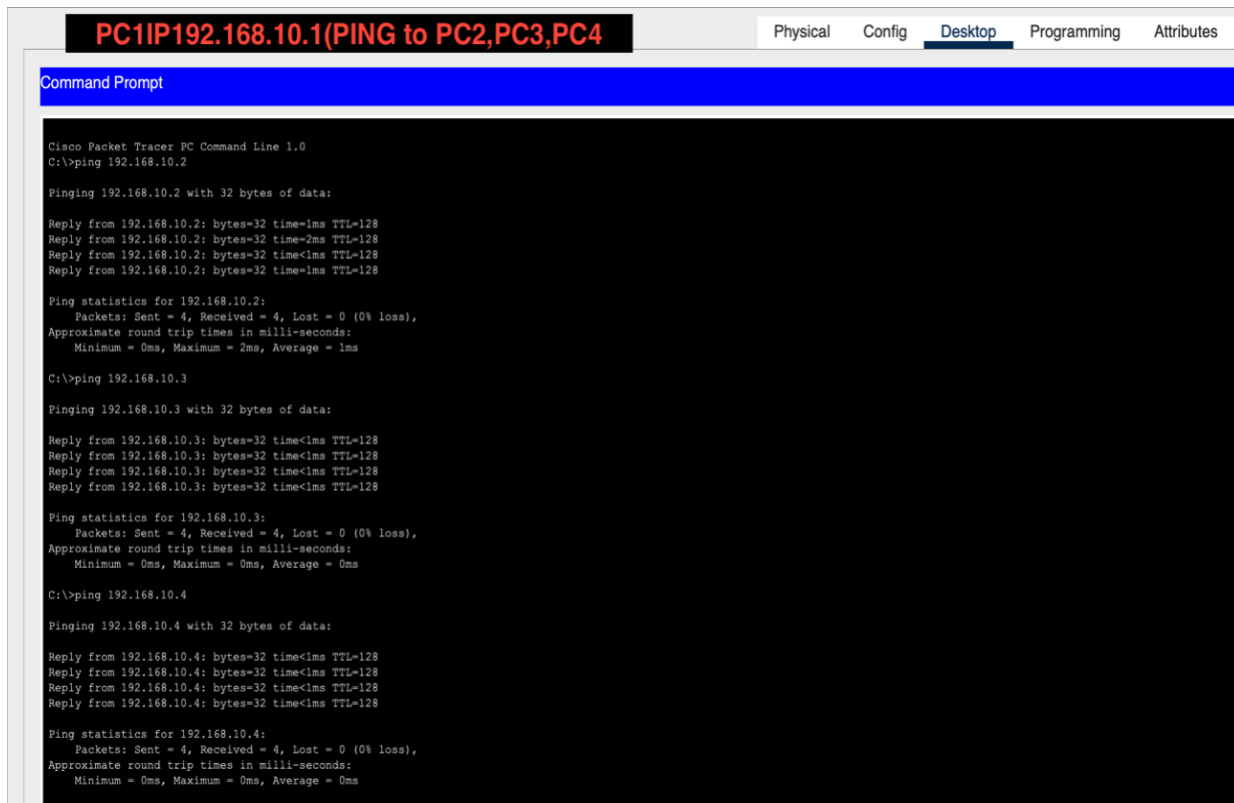
Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

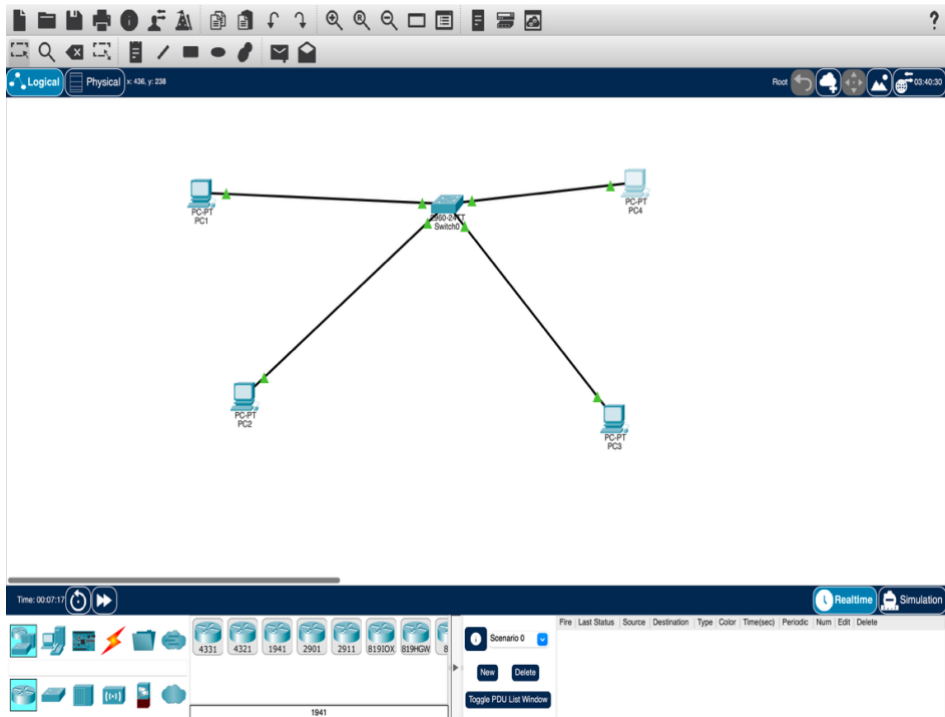
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

2.2 Subnetting the Network

- Figure 5: Topology



• Figure 6: Subnet IP Addresses

PC Subnet 1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface ☒ FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.1

Subnet Mask 255.255.255.128

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::202:17FF:FE1D:9388

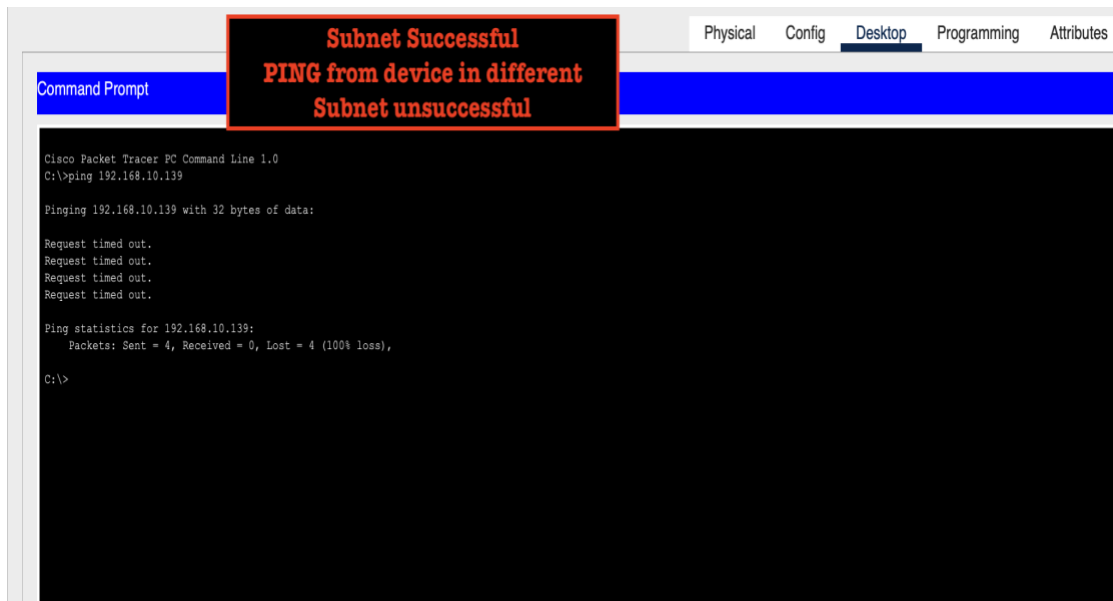
Default Gateway

DNS Server

PC Subnet 2

Physical	Config	Desktop	Programming	Attributes
IP Configuration				
Interface FastEthernet0				
IP Configuration				
☐ DHCP ☒ Static				
IPv4 Address 192.168.10.130				
Subnet Mask 255.255.255.128				
Default Gateway 0.0.0.0				
DNS Server 0.0.0.0				
IPv6 Configuration				
☐ Automatic ☒ Static				
IPv6 Address				
Link Local Address FE80::202:16FF:FE87:B95				
Default Gateway				
DNS Server				
802.1X				
☐ Use 802.1X Security				
Authentication MD5				
Username				
Password				

- **Figure 7: Ping Tests**



3. Task 3 VLANs (Basic Segmentation)

- In this task I configured VLAN 10 (Sales) and VLAN 20 (HR) on a 2960 switch with two devices assigned to each VLAN and the ports configured accordingly. The ping test confirmed that devices within the same VLAN could now communicate while devices across different VLANs could not, demonstrating successful Layer 2 segmentation.
- **Evidence**
- **Figure 8: VLAN Configuration**

- Evidence
- Figure 8: VLAN Configuration

```
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
enable
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10 Sales	active	Fa0/1, Fa0/2
20 HR	active	Fa0/3, Fa0/4
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
Switch(config)#
```

Figure 9 :Ping Test Results

PING FROM VLAN 10 DEVICE TO VLAN 20 DEVICE
Unsuccessful

PhysicalConfig**Desktop**ProgrammingAttributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PING ON VLAN 10 DEVICES on same network Successful

PhysicalConfig**Desktop**ProgrammingAttributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=2ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

4. Task 4 – Inter-VLAN Routing

- In this task I added a router and connected it the switch with a straight through cable, then configured using a router-on-a-stick method then created a sub interface for VLAN 10 and VLAN 20 using 802.1Q encapsulation then added gateways for each VLAN leading to communication between the networks to be established and successful ping between the VLANs.
- Evidence

Figure 10: Router Configuration

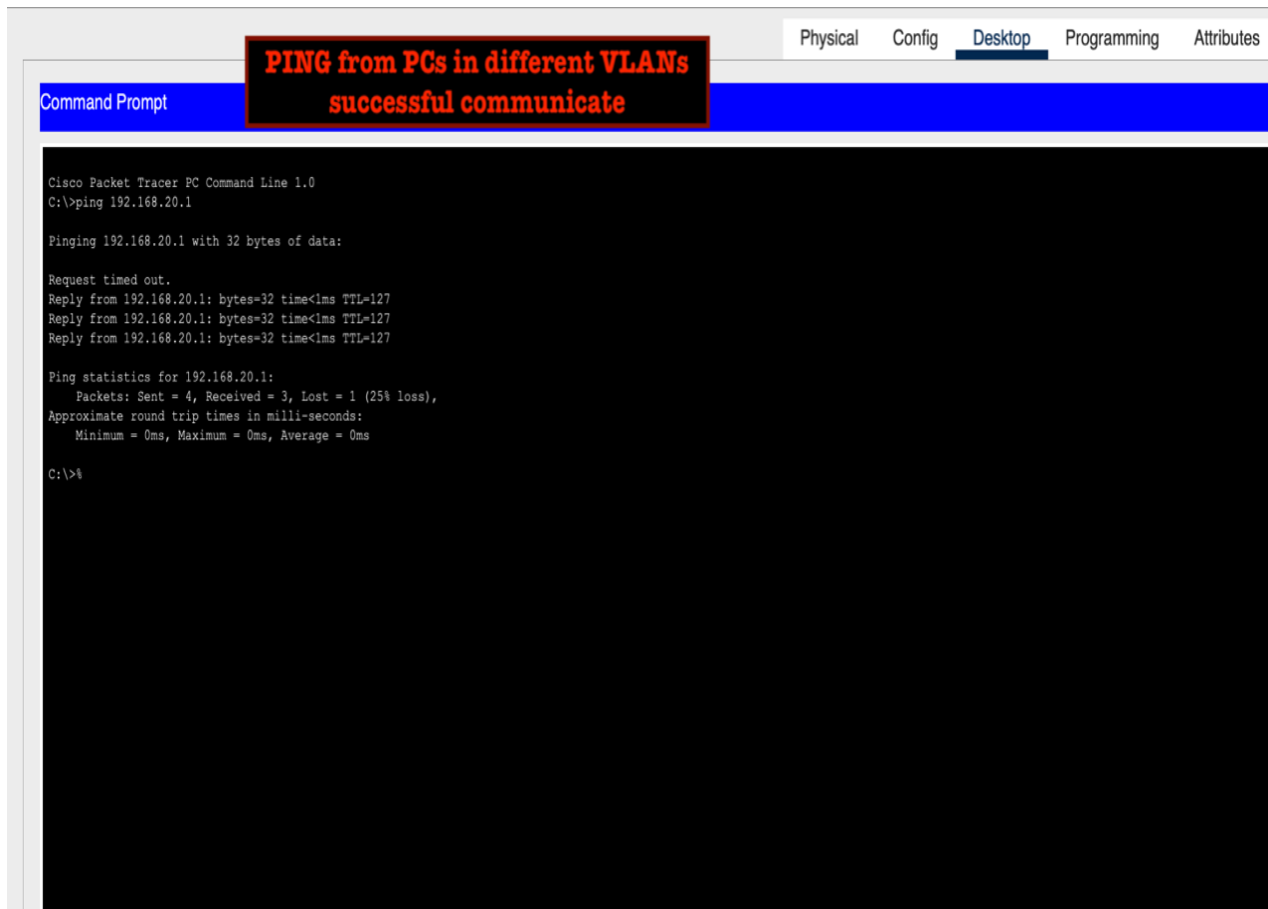
```
!
spanning-tree mode pvst
!
!
!
!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
!
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.254 255.255.255.0
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.254 255.255.255.0
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
shutdown
!
router rip
!
ip classless
!

Router#show ip interface brief
^
% Invalid input detected at '^' marker.

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES unset  up          up
GigabitEthernet0/0.10 192.168.10.254  YES manual  up          up
GigabitEthernet0/0.20 192.168.20.254  YES manual  up          up
GigabitEthernet0/1    unassigned      YES unset  administratively down down
Vlan1            unassigned      YES unset  administratively down down
Router#
```

router
configurations

- **Figure 11: Successful Inter-VLAN Ping**



5. Task 5 – Final Network Project

What I Built

In this project, I designed and configured a small network using Cisco Packet Tracer, building from basic network connectivity to a fully functioning and routed network.

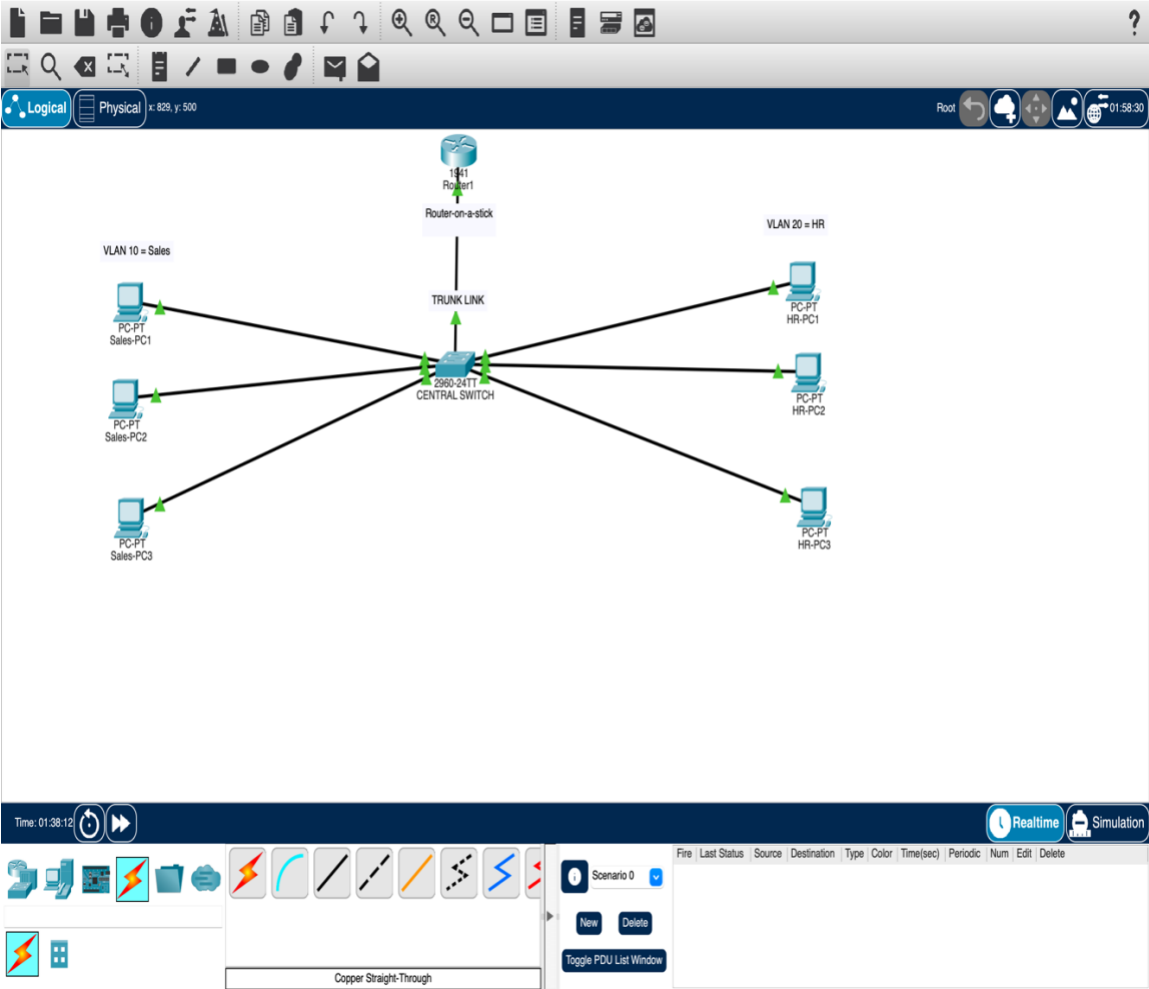
I started with creating a simple network consisting of two PCs connected through a single switch, then applied IP addressing and subnetting across four PCs, dividing a /24 network into two /25 subnets.

Then I implemented a VLAN segmentation by creating VLAN 10 and VLAN 20) on a single physical switch, creating isolation at the switch level then Finally I configured inter-VLAN routing using a Router-on-a-Stick method, with a trunk link carrying 802.1Q tagged traffic thus handling routing between VLANs.

What I Learned

Through this project I learned that networking is built in layers, physical connections, IP - addressing and routing each solve a different problem. I developed a deeper understanding of how a switch operate at IOS Layer 2 using MAC addresses, while routers operate at Layer 3 using IP addresses. I also learned that VLANs create virtual separation on one physical switch, and that Router-on-a-Stick is an efficient solution for inter-VLAN routing using a single cable and how subnetting is a crucial factor or method in networking.

Figure 12: Final Network Topology



- **Figure 13: Working Network Connectivity**

```
Switch>enable
Switch#show running-config
Building configuration...

Current configuration : 1385 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/2
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/3
switchport access vlan 20
!
interface FastEthernet0/4
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/5
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/6
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/7
```

REFERENCE

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